

Width [SimTalk] - FootPath

Sets the **Width** of the *Footpath* designated by <Path>.

Type

Attribute

Syntax

```
<Path>.Width:length
```

Assignment Value

You can assign a value of data type *length*.

Note

In *SimTalk* 2.0 you can specify the length units m, mm, km, cm, yd, ft, and in.

Type in the unit directly after the values, without a separating blank space, for example 10m or 10.2m.

You can specify the unit for *floating point values* and for *integer values*.

Example

```
MyFootPath.Width := 2 // meters
```

See also

[Width \[text box\] - FootPath](#)

WorkerPool

WorkerPool

Use the object *WorkerPool* for modeling the lounge or the staff room of your plant.

Image



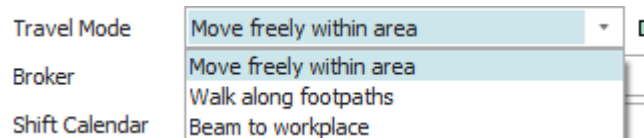
Description

The *Worker-WorkerPool-Workplace-FootPath*-concept refines the *Broker-Importer-Exporter*-concept. The *WorkerPool* represents the lounge or the staff room of your plant. The *Workers* are created in the **WorkerPool** and they stay there when they do not work and are waiting for an order.

As a rule you type in the **Names** of the *Workers*, which you want to use in your simulation model, into the **Workers to Create** table. You can also drag the *Worker* from the *Class Library* onto the icon of the *WorkerPool* and drop him there. The *WorkerPool* appends a line to the **Workers to Create** table for this *Worker*. When it already contains a line for this *Worker*, the *WorkerPool* increments the amount by 1.

You can also create individual *Workers*, which you do not want to generate via the **Workers to Create** table, with the method **create** for a single simulation run. This only works in the *WorkerPool* though. Be aware that *Plant Simulation* deletes this *Worker* when you reset the model and then only creates the *Workers* from the **Workers to Create** table at the next **Init** event that takes place.

You can select the **Travel Mode** with which the *Worker* gets to his work station:



The **Broker** brokers the *Worker* to the individual work stations.

Note

If no *Broker* exists when creating the *Workers*, *Plant Simulation* sets the attribute **AutomaticMediation** of the *Workers* to **false**.

As soon as a *Worker* can provide a service, the *Broker* sends him from his *WorkerPool* to the work station that ordered him:

- When the *Worker* walks freely within the area of the simulation model, he walks on the shortest route to the *WorkPlace* and walks around obstacles you designated.

The *Worker* consumes time, when he walks freely within the area to the *Workplace*.

- When you modeled the work station using one of the material flow objects and the *Workplace* and connected the *WorkerPool* and the *Workplace* with a *FootPath*, *Plant Simulation* animates the *Worker* while he moves on the *FootPath* and while he works on the *Workplace*.

The *Worker* consumes time, when he walks on the *FootPath*.

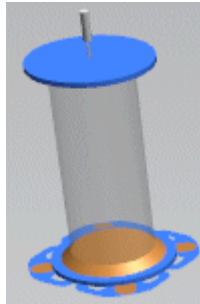
- If you activated **Beam to workplace** and you inserted a *Workplace*, and the *Worker*, who is supposed to work there, cannot reach it via a *FootPath*, *Plant Simulation* transports the *Worker* directly to the *Workplace* and animates him there.
- If you activated **Workers can work remotely** and if you did not insert a *Workplace*, if no *Workplace* exists that supports this service, or if all *Workplaces* that support this service are occupied, *Plant*

Simulation seemingly beams the *Worker* to the work station itself, where he will perform his task. It does not animate him there, but continues to animate him in the *WorkerPool*.

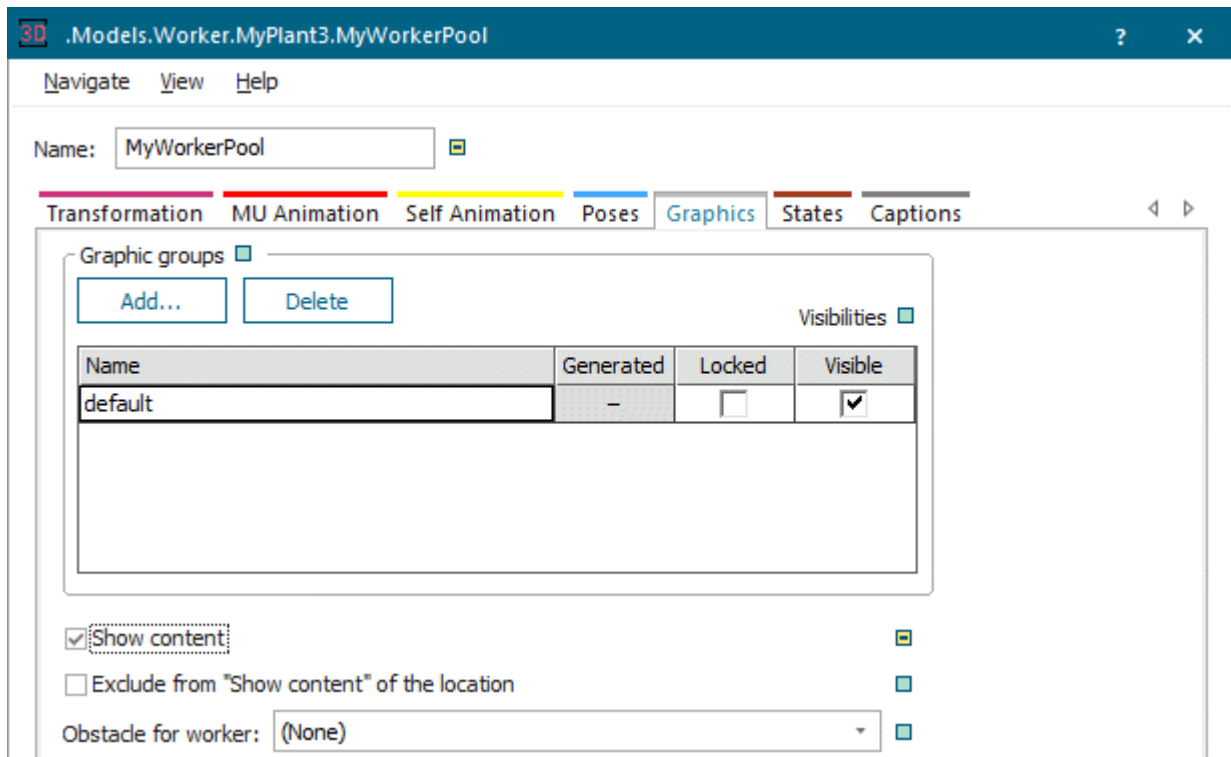
If your plant works in shifts, the *Worker* walks back to his *WorkerPool* at the end of his shift. He then deposits the parts he could not deliver into the **Parts Buffer**, which you assigned to the *WorkerPool*. The parts buffer can be any one of the material flow objects, which accepts parts. When you did not assign a parts buffer to the *WorkerPool*, *Plant Simulation* shows a warning and stops the simulation.

If you activated the *transport importer* in the *parts buffer*, one of the available *Workers* carries the parts in the *parts buffer* to the destination object, after the next shift has started. If you did not activate the *transport importer*, the parts in the *parts buffer* are moved on according to the **Exit Strategy** which you selected for the *parts buffer*.

The *WorkerPool* looks like this:



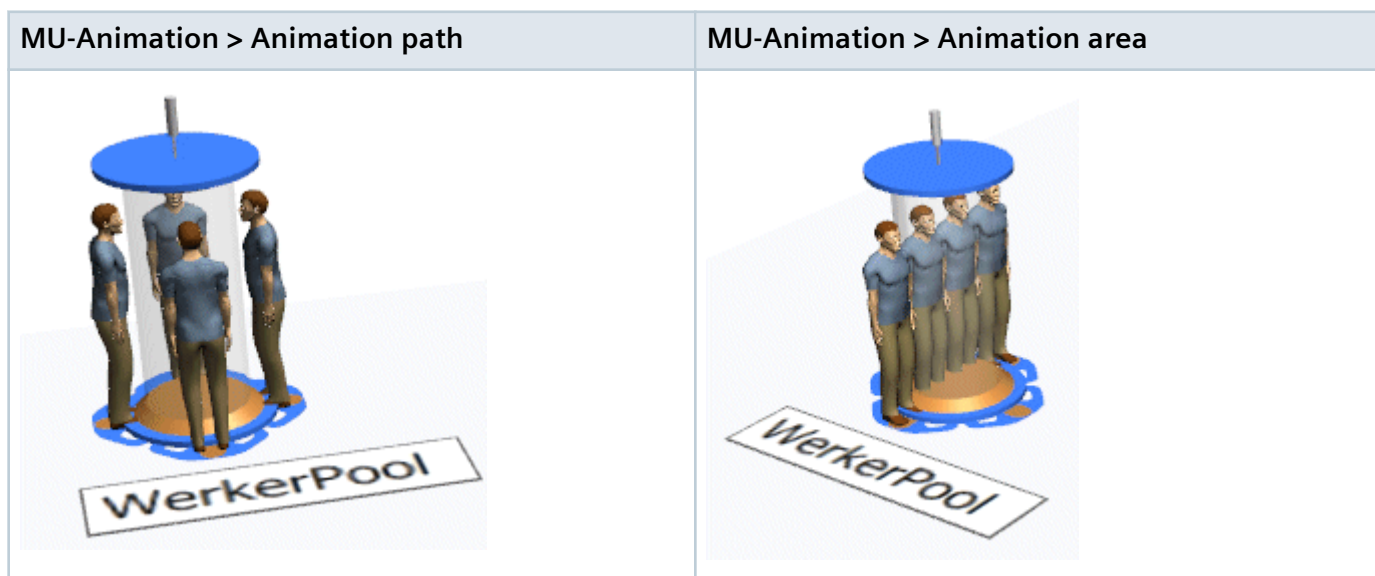
The *WorkerPool* initially does not show any *Worker* at all. To see the *Workers*, select the check box **Show Content** on the tab **Graphics** of the *WorkerPool*



Then click  **Start Simulation**.



On the tab **MU-Animation** you can select to show the *Workers* in the *WorkerPool* on an **Animation Path** or on an **animation area distributed along the Y-dimension**.



To show a **tooltip** with information about the *WorkerPool*, hover with the mouse over it.

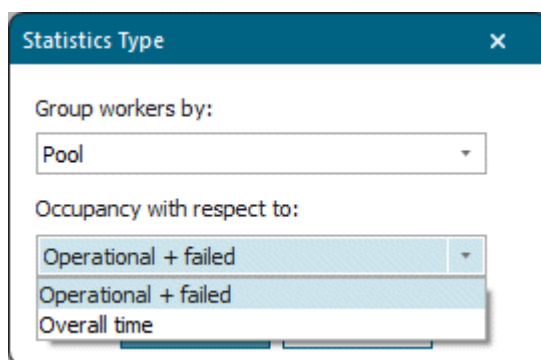
To change the length of the graphic and the anchor points of the *WorkerPool*, click **Show Manipulators**



on the **Edit** ribbon tab or press **M** on the keyboard.

Showing Statistics of the Workers in a Chart

To show statistics of the *Workers* managed by the *WorkerPool*, drag it over a *Chart* and drop it there. Select the settings that you need.

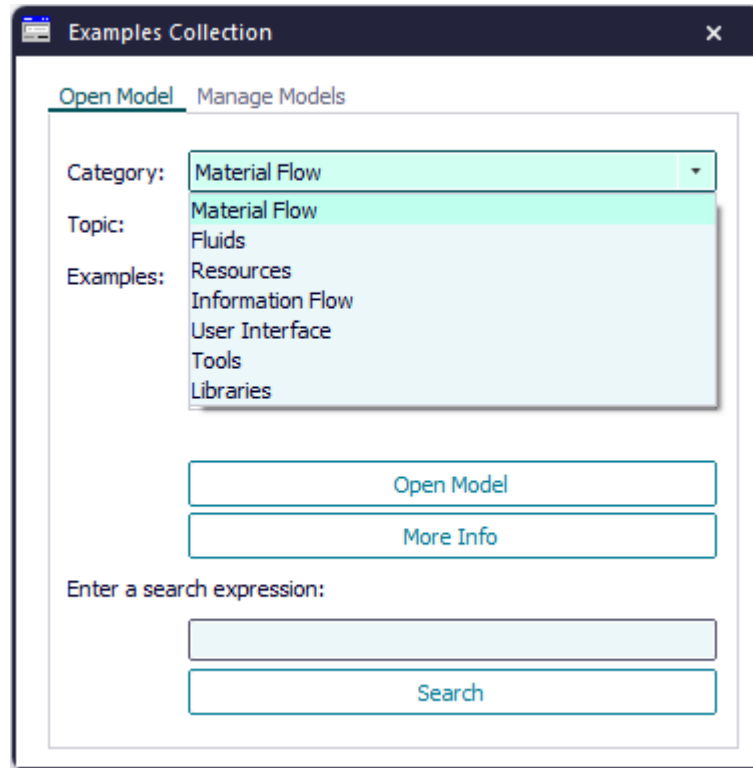


Add the Object to the Simulation Model



To add the object *WorkerPool* to your simulation model, click **Manage Class Library > Basic Objects > Resources > WorkerPool** on the **Home** ribbon tab.

Compare the sample models: Click the **Window** ribbon tab, click **Start Page > Getting Started > Example Models > Small Examples**. Then, select the respective **Category**, the **Topic**, and the **Example** in the dialog **Examples Collection**, and click **Open Model**.



See also

[Show Worker Statistics in a Chart](#)

[Model Workers and the Jobs They Do](#)

[Worker-WorkerPool-Workplace-FootPath-concept](#)

[Broker-Importer-Exporter-concept](#)

Dialog Box of the WorkerPool

Dialog Box of the WorkerPool

Double-click the icon of the *WorkerPool* to open its dialog box.

Edit Simulation Properties

In the dialog box you can change the **simulation properties** of the object. The shared properties are described under [Dialog Items of the Objects](#).

Edit Animation Properties

To edit the **3D properties** of the object in the dialog box [Edit 3D Properties](#):

- Click the button  **Edit 3D Properties** in the lower left corner of the simulation properties dialog box.
- Select the object in the model and press the **spacebar**.

To manipulate the graphic of the object, click [Show Manipulators](#) on the **Edit** ribbon tab or press **M** on the keyboard.

Tab Attributes

Tab Attributes

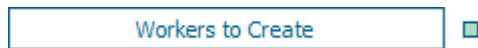
The tab **Attributes** provides the settings, which the object offers.

The settings are listed in the table of contents to the left.

The shared properties are described under the [Tab Attributes](#).

Workers to Create

To open the table, into which you type information pertaining to the *Workers* to be created and inserted into your model, click this button.



Remarks

Note

Plant Simulation always resolves **relative paths** in relation to the *Frame* in which the *WorkerPool* is inserted. If *Plant Simulation* cannot resolve a path in the **Scope** array during the simulation, it shows an error message and asks if you would like to stop the simulation.

This also applies to the **Home Location**.

Click the  button to deactivate **Inheritance** . Only then you can enter data into the table.

	Worker	Amount	Shift	Speed	Efficiency	Home Location	Scope	Additional Services
1	*.Resources.Worker	1				Home1	Station1; Station2	-
2	*.Resources.Worker	1				Home11	Station11; Station21	

Buttons: OK, Cancel, Apply

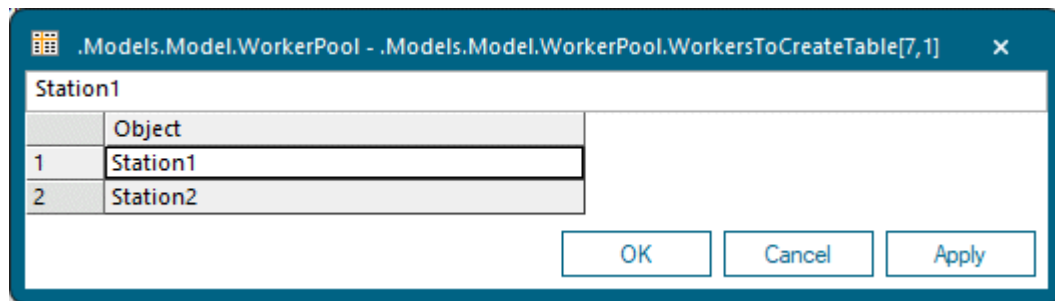
Proceed as follows:

- Type the class of the *Worker*, which the *WorkerPool* uses as a template for the *Workers* to be created, into the cells below **Worker**. You might, for example, type in `.Resources.Worker` or `.UserObjects.MyWorker`.
- Type the number of the *Workers* to be created into the cells below **Amount**.
- Type in the name of a **Shift** which you defined in the associated **Shift Calendar [WorkerPool]**.
- If you want the *Worker* to walk with a different **Speed** than the one defined in its class, type this speed into the cell below **Speed**. This deactivates inheritance for the speed, i.e., speed changes in the class will not affect the *Workers* defined in this row. If you do not type in anything, the *Worker* walks with the speed specified in the class, meaning he inherits that value from the class.
- Type in the **Efficiency** of the *Workers* to be created into the cells below **Efficiency**. You might, for example, enter 80 for an efficiency of 80 percent.
- Type in the **Home Location**, i.e., the path to the *Workplace* to which the *Worker* walks, once he has finished his current job.

- Type in the **Scope**, i.e., the path to the material flow objects to which the *Worker* can be brokered. The scope also encompasses *sub-Frames*.

Proceed as follows:

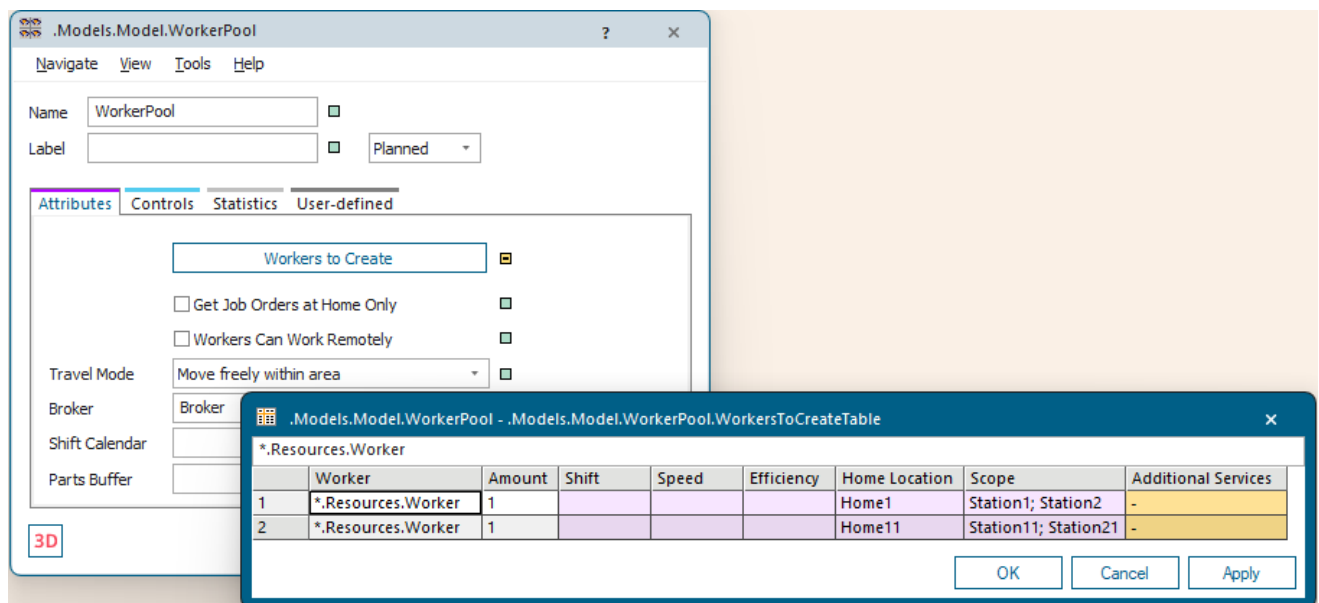
- Click in the cell **Scope** of the *Worker* whose scope you want to set.
- This opens a table with one empty cell.
- Select the object in the *Frame* to which the *Worker* can be brokered and drop it into the cell of the table.



Press **Enter** to add another cell and drag the next object to this new cell.

Repeat this as often as necessary.

- Click **OK** in the table. *Plant Simulation* then inserts the objects you added to the table into the cell **Scope** of the **Workers to Create** table and separates them with a semicolon.



- Click **OK** in the window of the **Workers to Create** table.
- Type the services which the *Worker* can provide, in addition to those you entered into the **Services** list, into the cells below **Additional Services**.

You can type different types of additional services into the columns to the right of **Additional Services**. You can also leave gaps between the services in the table. This way you can type in identical services one below the other, even if you want some *Workers* to not provide these services.

Note

The names of the services are not case-sensitive. The *WorkerPool* attempts to export the services in the order in which you entered them into the table.

Note

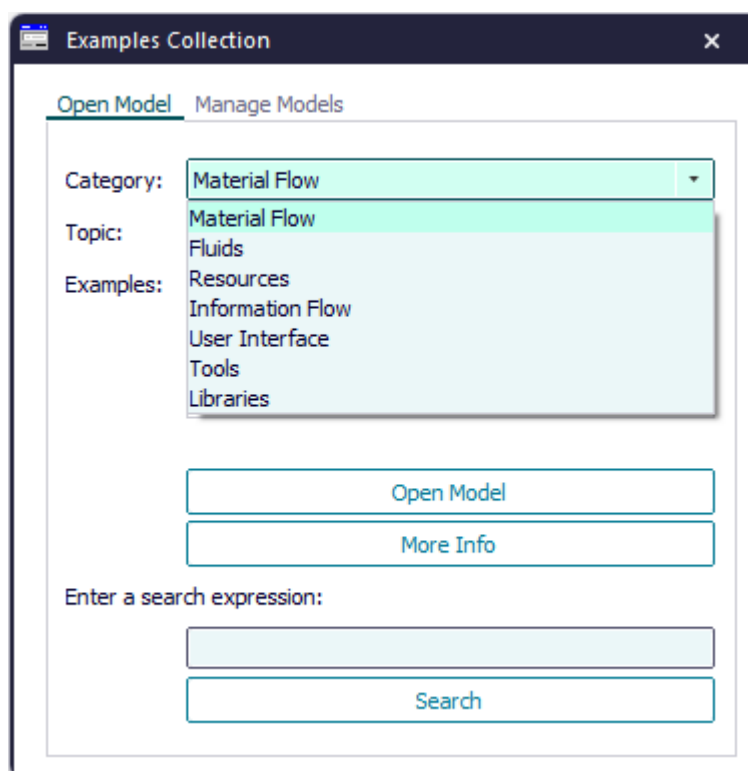
To save memory and improve access speed, all places which use such a case-insensitive string point to the same string in main memory. The visible and unexpected result is that the first occurrence of the string defines how the string is written in terms of upper- and lower-casing.

In *SimTalk* you can compare if strings are case-insensitive with the `~=` operator, compare **Relational Operators**. The change in the **Workers to Create** table only affects the simulation run if the change takes place before the initialization.

Plant Simulation enters the services you defined here into the list of offered services of the **Broker**: To view them, select **View > Offered Services**.

You can also change the **Workers to Create** table within the **Init Control** of the *WorkerPool* for the upcoming simulation run. This is not possible in an **init** method, as the method is executed too late in time.

Compare the sample models: Click the **Window** ribbon tab, click **Start Page > Getting Started > Example Models > Small Examples**. Then, select the respective **Category**, the **Topic**, and the **Example** in the dialog **Examples Collection**, and click **Open Model**.



SimTalk

[AdditionalServices \[SimTalk\]](#)

[Amount \[SimTalk\] - WorkerPool](#)

[Efficiency \[SimTalk\] - WorkerPool](#)

[getWorkersToCreateTable \[SimTalk\]](#)

[setWorkersToCreateTable \[SimTalk\]](#)

[setServices \[SimTalk\] - importer](#)

[Shift \[SimTalk\] - WorkerPool](#)

[Speed \[SimTalk\] - WorkerPool](#)

[Worker \[SimTalk\] - WorkerPool](#)

See also

[How the Worker Decides Where to Work](#)

Set Home Location and Scope for the Worker

Choose the Nearest Worker [check box]

Tab Scope

Set How Many Workers Are Created During Initialization

Init Control [WorkerPool]

Context Menu of Embedded Lists

GetJobOrdersAtHomeOnly [SimTalk]

Video on YouTube

<https://youtu.be/HiPziA8kxc0?si=qYFCTtAc7V4GzXjQ&t=178>

Get Job Orders At Home Only

To make the *Worker* get new job orders for performing a service at a station in the *WorkerPool* or at his **Home Location** only, select this check box.

☐ Get Job Orders at Home Only ☐

Remarks

A *Worker*, who is staying on a *Workplace*, for which the setting **Worker Stays Here After Completing the Job** is activated, cannot be brokered in this case. You thus have to clear **Worker Stays Here After Completing the Job** in all *Workplaces* so that brokering of the *Workers* works correctly.

Clear the check box, to allow the *Worker* to get his job orders anywhere, no matter if he walks freely in the area or if he walks on a *FootPath*.

SimTalk

[GetJobOrdersAtHomeOnly \[SimTalk\]](#)

See also

[Home Location](#)

[Worker Stays Here After Completing the Job](#)

[WorkerStaysHere \[SimTalk\]](#)

Workers Can Work Remotely

To make the *Worker* do his job from his current location, when there is no free *Workplace* available at the station, to which he was assigned, select this check box.

☐ Workers Can Work Remotely ☒

Remarks

You can use this feature to build a model, which does not contain any *Workplaces*. The *Worker* will then stay in the *WorkerPool* the entire time and work from there.

When a free *Workplace* is available, the *Worker* will use this *Workplace*, even if you select **Workers Can Work Remotely**. If you want to prevent a certain service from using a *Workplace*, you can either use an *Exporter* or restrict the *Workplaces* to certain services.

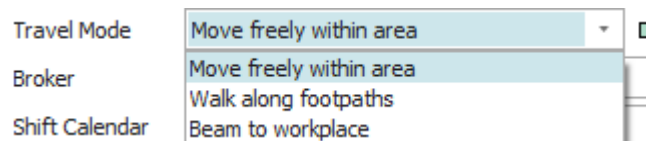
SimTalk

[WorkersCanWorkRemotely \[SimTalk\]](#)

Travel Mode

Travel Mode [drop-down list]

Select the **Travel Mode** of the *Worker* from the drop-down list.



Remarks

You can select one of these settings:

- **Move freely within area**

The *Worker* walks freely within the area of the simulation model and walks around **Obstacles**, which you defined in 3D.

[Model a Worker Who Walks Freely Within the Model](#) gives a short introduction to the topic.

- **Walk along footpaths**

The *Worker* walks on the *Footpaths*, which you inserted between the *Stations/Workplaces*, to the station at which he is going to work.

- **Beam to workplace**

The *Worker* is beamed (teleported) to the *Workplace* to which he was brokered and at which he is going to work instantly, without using up any time at all, provided he cannot get to this *Workplace* on any *Footpath*. In this case the *Worker* functions as an **Exporter** with a capacity of 1.

You can use **Beam to workplace** to build a model which does not contain any *FootPaths*. The *Worker* then always jumps between the *WorkerPool* and the *Workplaces*.

Note

The **Travel Mode** is only transmitted from the *WorkerPool* to the *Workers* when they are created. The **Travel Mode** will not be changed after the *Workers* are created.

SimTalk**WorkersTravelMode [SimTalk]**

See also

Set How Many Workers Are Created During Initialization**Model a Worker Who Walks Freely Within the Model**

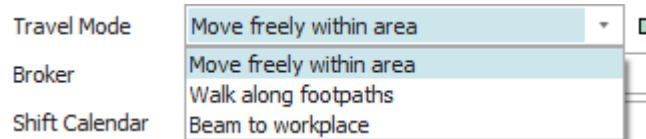
Edit 3D Properties [dialog] > Graphics [tab] > Obstacle for the Worker [drop-down list] in 3D

Edit 3D Properties [dialog] > Graphic Settings in 3D

Barred Area in 3D

Move freely within area

Makes the *Worker* walk freely within the area of the simulation model and walk around the **obstacles**, which you defined.



Remarks

Plant Simulation automatically computes the shortest route between the *WorkPlaces* and between the *WorkerPool* and the *WorkPlaces*. It uses the **layout of the model** to compute the shortest route to a *WorkPlace*.

Note

We use the 3D layout to compute the route of the *Worker*. It normally represents the looks of your facility close to reality, meaning that the position, the dimensions, and the orientation of the machines and the fixtures of the building closely match the real plant.

To compute the shortest route, *Plant Simulation* uses information about those areas of the model, which the *Worker* cannot walk on or cross. These areas encompass all kinds of obstacles, for example walls, pillars, machines, conveyors, safety fences, etc. *Plant Simulation* automatically detects these areas. By default, we marked the *material flow objects*, the *animatable objects*, and the *graphics*, which belong to the *Frame*, as **Barred Areas**. *Information flow objects* and *user interface objects* by default are not **Barred Areas**. You can set if an object as an **obstacle** for the *Worker* under **Edit 3D Properties > Tab Graphics > Obstacle for the Worker**. For graphics, you can do this under **Edit 3D Properties > Tab Graphic Settings**, for example to allow the *Worker* to cross a *Track*. *MUs* do not provide this setting.



All settings, except for **None** automatically create a barred area.

To designate an area in the plant, which the *Worker* can enter, but should not, for example the pivoting area of a crane, you can manually create a **Barred Area**.

To allow the *Worker* to walk around or through obstacles, use a *FootPath*. Let's say that you want your *Worker* to **walk through a door** which in part of a wall that you defined as an obstacle. Then you can lay a *FootPath* through the door on which he can walk through it. The *FootPath* always has to be a little longer than the depth of the wall.

For the *FootPath* not the length, which you type in on the tab **Attributes** is decisive, but the **Length** of the *FootPath* in the scene. The distance covered of the *Worker* is determined by the starting point and the end point of the *FootPath* as well by the anchor points.

If the *FootPath* includes an **incline**, the *Worker* moves slower on the incline than on a flat surface. *Plant Simulation* achieves this by extending the imagined distance for the incline segments. When computing the length of an inclined segment, *Plant Simulation* triples the height difference.

Compare this example: A segment of a *FootPath* has a dimension of 1 m in the x-direction and 1 m in the z-direction, thus actually is 1.414 m long. The imagined dimension however is 1 m in the x-direction and 3 m in z-direction, so that *Plant Simulation* computes an effective length of 3.162 m.

The *Worker* always walks on the route, which has the shortest effective length and which he can thus cover in the shortest time. This can cause the *Worker* to take an actually longer but flatter route to the destination instead of a shorter route which contains inclines and gradients.

For computing the effective route length *Plant Simulation* triples the height difference. The resulting speeds of the *Worker* are realistic in general. The speed of the *Worker* decreases to a third, when the *Worker* moves vertically up or down, for example when climbing a **ladder**. The speed decreases to about half, when the *Worker* negotiates an incline of 38 degrees, for example a when walking on a **stairway**.

Note

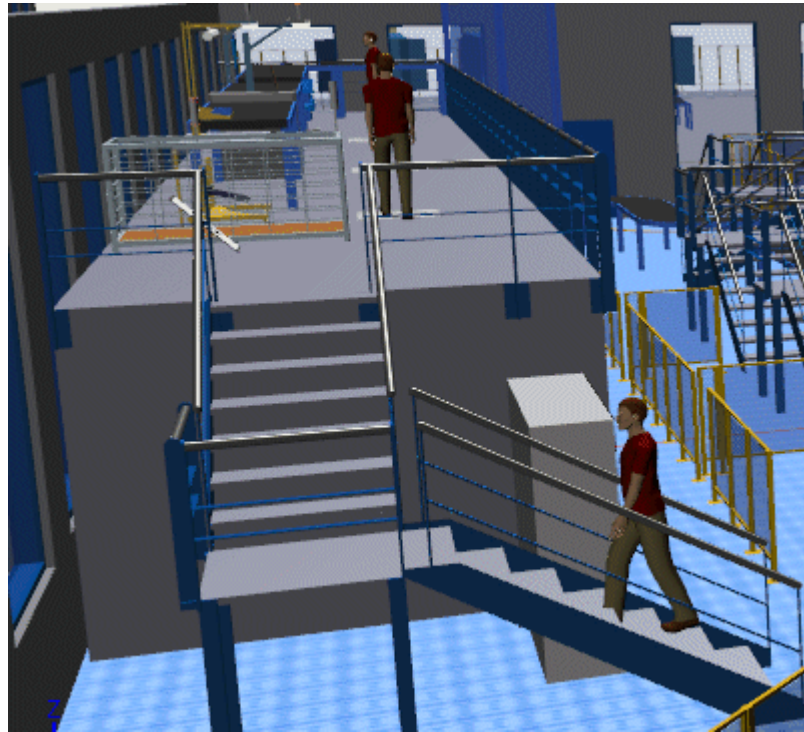
The *Worker* walks slower on an incline than he walks on a flat surface.

Note

When the *Worker* **walks freely within the area**, he can only get to a different floor in the building by walking up or down a ramp on a *Footpath* or walking up and down **Stairs**.

The example model **Factory 51** demonstrates how you can use **Stairways**.

Move freely within area



SimTalk

[WorkersTravelMode \[SimTalk\]](#)

See also

[Obstacle for the Worker \[drop-down list\]](#)

[Model a Worker Who Waits in Line](#)

[Model a Worker Who Walks Freely Within the Model](#)

[Set How Many Workers Are Created During Initialization](#)

[How the Worker Decides Where to Work](#)

[Set Home Location and Scope for the Worker](#)

[Choose the Nearest Worker \[check box\]](#)

[Edit 3D Properties > Graphics \[tab\] > Worker](#)

[Edit 3D Properties > Graphic Settings](#)

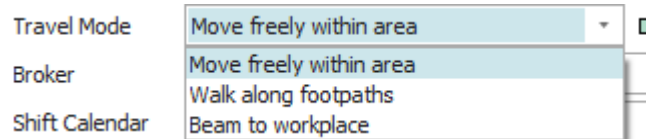
[Barred Area in 3D](#)

Back to

Travel Mode [drop-down list]

Walk along footpaths

Makes the *Worker* walk on the *Footpaths*, which you inserted between the *Stations/Workplaces*, to the station at which he is going to work.



Remarks

To model and to simulate the distances between the *WorkerPool* and the work stations or between the work stations, insert *FootPaths*. *Plant Simulation* animates the *Workers* on the *FootPaths* when they walk toward the work stations or when they walk between the stations.

You can link several *FootPaths* with *Connectors* to create a network of *FootPaths*. The *Worker* walks within this network from his *WorkerPool* to the *Workplace*. Provided the *WorkerPool* and the *Workplace* are connected to the same network of *FootPaths*, the *Worker* walks from the *WorkerPool* to the *Workplace* and back on the shortest possible route.

A network of *FootPaths* consists of all *FootPaths* that are joined with *Connectors*, as well as the *WorkerPools* and the *Workplaces* that are joined with one of these *FootPaths* with a *Connector*.

The time it takes the *Worker* to walk from his *WorkerPool* to the *Workplace* depends on his **Speed** and the combined **Lengths** of the *FootPaths* he covers to reach his destination. *Plant Simulation* recognizes on which *FootPaths* the *Worker* walks if he exits a *FootPath* at another end than the one at which he entered it. If the *Worker* does not walk to his destination on a *FootPath* or if he is beamed to his destination, he does not consume any time at all.

SimTalk

[WorkersTravelMode \[SimTalk\]](#)

See also

[Set How Many Workers Are Created During Initialization](#)

[Edit 3D Properties > Tab Graphics > Obstacle for the Worker](#)

[Edit 3D Properties > Tab Graphic Settings](#)

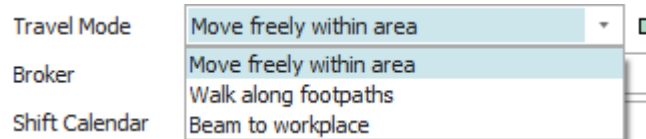
[Barred Area](#)

Back to

Travel Mode [drop-down list]

Beam to workplace

Beams (teleports) the *Worker* to the *Workplace* to which he was brokered and at which he is going to work instantly, without using up any time at all, provided he cannot get to this *Workplace* on any *Footpath*.



Remarks

For the setting **Beam to workplace** the *Worker* functions as an *Exporter* with a capacity of 1.

You can use **Beam to workplace** to build a model which does not contain any *FootPaths*. The *Worker* then always jumps between the *WorkerPool* and the *Workplaces*.

SimTalk

[WorkersTravelMode \[SimTalk\]](#)

[teleportTo \[SimTalk\]](#)

[teleportToHome \[SimTalk\]](#)

[teleportToPool \[SimTalk\]](#)

See also

[Exporter \[object\]](#)

[Set How Many Workers Are Created During Initialization](#)

[Model a Worker Who Walks Freely Within the Model](#)

[Edit 3D Properties > Tab Graphics > Obstacle for the Worker](#)

[Edit 3D Properties > Tab Graphic Settings](#)

[Barred Area](#)

[Back to](#)

[Travel Mode \[drop-down list\]](#)

Broker [WorkerPool]

Select the *Broker*, who assigns the services to the *Worker*.

Broker: ... 

Remarks

Click the ellipsis button and select the *Broker*, who assigns the services to the *Worker*, in the dialog **Select Object**.

Or type in the path to and the name of the *Broker*. You might, for example, type in `.Models.Model.MyBroker`.

Or select the *Broker* in a *Frame*, drag it to the text box and drop it there.

Note

To open the dialog of the object, whose name you typed into the text box, press **F2**.

SimTalk

BrokerPath [SimTalk] - WorkerPool

See also

Broker [object]

Assigned Workers [WorkerPool]

Select Object [for controls]

Shift Calendar [WorkerPool]

Select the *ShiftCalendar*. It contains the data of the shifts in your installation and controls during which shifts the *WorkerPool* works.

Shift Calendar 

Remarks

Click the ellipsis button and select the *ShiftCalendar* in the dialog **Select Object**.

Instead of clicking the ellipsis button, you can also select the *ShiftCalendar* in a *Frame*, drag it to the text box, and drop it there.

This automatically enters the object into the list of **Objects** on the tab **Resources** of the *ShiftCalendar*.

If the *Workers*, which the *WorkerPool* manages, works in shifts, you also have to select the **Parts Buffer**. The *Worker* deposits the parts, which he could not deliver, onto this object, when he walks back to the *WorkerPool* at the end of his shift.

SimTalk

[ShiftCalendarObject \[SimTalk\] - material flow objects](#)

See also

[ShiftCalendar \[object\]](#)

[Select Object \[for controls\]](#)

Parts Buffer

Select the **Parts Buffer**. This is the object into which the *Worker* deposits the parts he could not deliver, when he walks back to the *WorkerPool* at the end of his shift.

Parts Buffer

Remarks

Click the ellipsis button and select the **Parts Buffer** in the dialog **Select Object**.

You can also drag a material flow object onto the image of the *WorkerPool* in your model *Frame* and drop it there to use it as the **Parts Buffer**.

The **Parts Buffer** can be any one of the material flow objects, which accepts parts. If you did not assign a parts buffer to the *WorkerPool*, and a *Worker* arrives with parts at the end of its shift in the *WorkerPool*, *Plant Simulation* shows a warning and stops the simulation.

If you activated the *transport importer* of the **Parts Buffer**, one of the available *Workers* carries the parts in the **Parts Buffer** to the destination object, after the next shift has started. If you did not activate the *transport importer*, the parts in the **Parts Buffer** are moved on according to the **Exit Strategy** which you selected for the **Parts Buffer**.

Note

This setting only applies, when you control the shifts worked in your facility with the [Shift Calendar \[WorkerPool\]](#).

SimTalk

[PartsBuffer \[SimTalk\]](#)

Tab Statistics [WorkerPool]

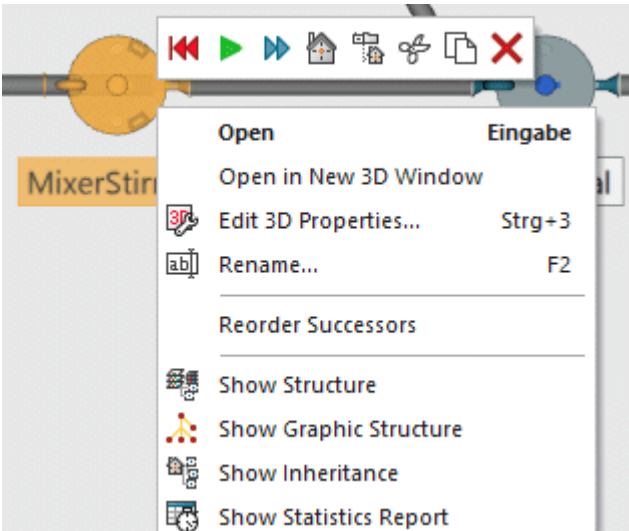
Plant Simulation shows the most important statistics values on the tab **Statistics**.

Item English	Description	Read-only attribute	Item German
Paused	Shows the portion of the statistics collection period during which the <i>WorkerPool</i> was paused	StatPausingCount [SimTalk]	Pausiert
Unplanned	Shows the portion of the statistics collection period during which the <i>WorkerPool</i> was unplanned, i.e., is not scheduled to work to the statistics collection period	StatUnplannedPortion [SimTalk]	Ungeplant
Average Traveled Distance	Shows the average distance in meters which the <i>Worker</i> traveled from the <i>WorkerPool</i> to the <i>Workplaces</i> attached to the stations and between the stations	StatAverageTraveledDistance [SimTalk] - WorkerPool	Durchschnittlich zurückgelegte Wegstrecke

For information about the other values on this tab, compare the [Tab Statistics](#) of the *Worker*.

To view [Resource Statistics of Stationary Resources](#) in the [Statistics Report](#), select **View > Show Statistics Report** in the dialog of the object.

You can also click the right mouse button in the *Frame* and select [Show Statistics Report](#) or you can press **F6**.



See also

[Resource Statistics \[check box\]](#)

Resource Type

Tab Statistics [Worker]

Tab Controls

Tab Controls [WorkerPool]

Click the ellipsis button and select a *Method* in the dialog **Select Object**. Type in the source code of the respective control.

Select the Path to an Existing Method

Click the ellipsis button. Navigate to the location of the *Method* in the dialog **Select Object [for controls]** and click **OK**. This inserts the name of the *Method* into the text box of the **Control**.



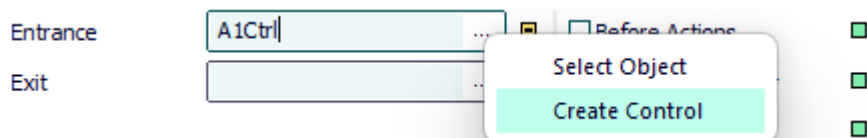
Press **F2** in the text box to open the *Method*. Then type in the source code of the **Control**.

Instead of choosing **Select Object**, you can also select the *Method* in a *Frame*, drag it to the text box and drop it there.

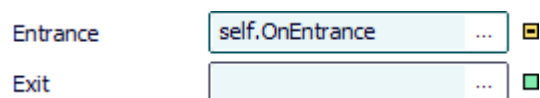
Create a Control That is a Method of the Object

Proceed as follows to create a control as a *user-defined attribute* of data type *Method*:

- Type a meaningful name into the text box and select **Create Control [context menu]**. *Plant Simulation* then inserts `self.Name_you_typed_in_for_the_control`, such as `self.A1Ctrl`.



- Select **Create Control** on the empty text box. *Plant Simulation* then inserts `self.OnBuilt_in_name_of_the_control`, such as `self.OnEntrance`.



Type the source code of this control into the *Method* that opens.

To edit the source code later on:

- Press **F2**.
- Or hold down **Shift** and double-click into the text box.
- Or select **Open Object** on the context menu.
- Or click the tab **User-defined** and double-click the name of the *Method* in the list.
- To delete this control, delete the *user-defined attribute*. If you only delete the name from the text box, the *user-defined attribute* is retained.

See also

[Entrance Control \[WorkerPool\]](#)

[Exit Control \[WorkerPool\]](#)

Entrance Control [WorkerPool]

Modifies the built-in behavior of the object. The object calls the **Entrance Control** as soon as the *Worker* has entered the *WorkerPool*.



Select the Path to an Existing Method

Click the ellipsis button. Navigate to the location of the *Method* in the dialog [Select Object \[for controls\]](#) and click **OK**. This inserts the name of the *Method* into the text box of the **Control**.



Press **F2** in the text box to open the *Method*. Then type in the source code of the **Control**.

Instead of choosing **Select Object**, you can also select the *Method* in a *Frame*, drag it to the text box and drop it there.

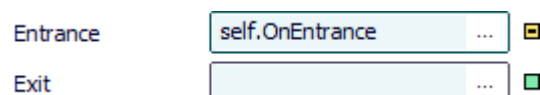
Create a Control That is a Method of the Object

Proceed as follows to create a control as a *user-defined attribute* of data type *Method*:

- Type a meaningful name into the text box and select [Create Control \[context menu\]](#). *Plant Simulation* then inserts `self.Name_you_typed_in_for_the_control`, such as `self.A1Ctrl`.



- Select **Create Control** on the empty text box. *Plant Simulation* then inserts `self.OnBuilt_in_name_of_the_control`, such as `self.OnEntrance`.



Type the source code of this control into the *Method* that opens.

To edit the source code later on:

- Press **F2**.

- Or hold down **Shift** and double-click into the text box.
- Or select **Open Object** on the context menu.
- Or click the tab **User-defined** and double-click the name of the *Method* in the list.
- To delete this control, delete the *user-defined attribute*. If you only delete the name from the text box, the *user-defined attribute* is retained.

SimTalk

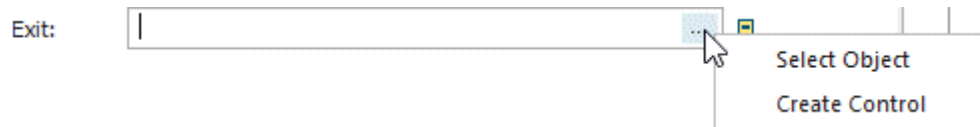
EntranceCtrl [SimTalk] - WorkerPool

See also

Select Object [for controls]

Exit Control [WorkerPool]

Modifies the built-in behavior of the object. The object calls the **Exit Control** as soon as the *Worker* has exited the *WorkerPool*.



Select the Path to an Existing Method

Click the ellipsis button. Navigate to the location of the *Method* in the dialog [Select Object \[for controls\]](#) and click **OK**. This inserts the name of the *Method* into the text box of the **Control**.



Press **F2** in the text box to open the *Method*. Then type in the source code of the **Control**.

Instead of choosing **Select Object**, you can also select the *Method* in a *Frame*, drag it to the text box and drop it there.

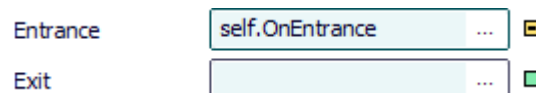
Create a Control That is a Method of the Object

Proceed as follows to create a control as a *user-defined attribute* of data type *Method*:

- Type a meaningful name into the text box and select [Create Control \[context menu\]](#). *Plant Simulation* then inserts `self.Name_you_typed_in_for_the_control`, such as `self.A1Ctrl`.



- Select **Create Control** on the empty text box. *Plant Simulation* then inserts `self.OnBuilt_in_name_of_the_control`, such as `self.OnEntrance`.



Type the source code of this control into the *Method* that opens.

To edit the source code later on:

- Press **F2**.

- Or hold down **Shift** and double-click into the text box.
- Or select **Open Object** on the context menu.
- Or click the tab **User-defined** and double-click the name of the *Method* in the list.
- To delete this control, delete the *user-defined attribute*. If you only delete the name from the text box, the *user-defined attribute* is retained.

SimTalk

ExitCtrl [SimTalk] - WorkerPool

See also

Select Object [for controls]

Tab User-defined

Define your own attributes as described under the [Tab User-defined](#).

Navigate Menu

The commands are described under the [Navigate Menu](#).

View Menu

View Menu [WorkerPool]

The **View Menu** provides commands to access its functions.

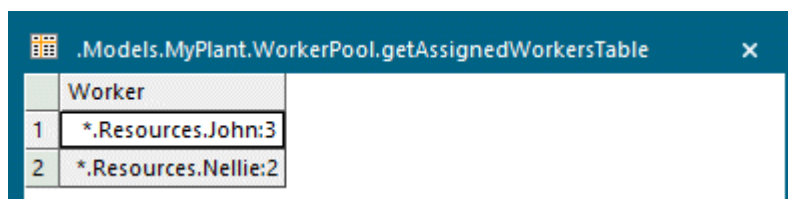
Refresh [on View menu]	Working Workers [WorkerPool]
Show Statistics Report [on View menu]	Associated Lockout Zones
Show Attributes and Methods [on View menu]	Associated Shift Calendar
Assigned Workers [WorkerPool]	Working Workers [WorkerPool]
Refresh [on View menu]	

See also

[View Menu \[general description\]](#)

Assigned Workers [WorkerPool]

Opens a list, which shows all *Workers* for which the *WorkerPool* provides services.



	Worker
1	*.Resources.John:3
2	*.Resources.Nellie:2

SimTalk

[getAssignedWorkersTable \[SimTalk\]](#)

[getAssignedWorker \[SimTalk\]](#)

[NumAssignedWorkers \[SimTalk\]](#)

See also

[Broker \[WorkerPool\]](#)

Working Workers [WorkerPool]

Opens a list, which shows information pertaining to the working workers.

.Models.MyPlant.WorkerPool.WorkingWorkers				
	Worker	Location	Target	Service
1	.Resources.Nellie:2	.Models.MyPlant.Workplace1	.Models.MyPlant.Processing2	StandardService

Remarks

The list shows:

- The **Worker** who provides a service or carries parts.
- The **Location**, i.e., the object on which the *Worker* is located at the moment. For a *Worker* walking freely in the area, the *WorkerPool* is shown here.
- The **Target** station at which the *Worker* is to provide the service or carry the part to.
- The **Service** which the *Worker* currently provides. If the *Worker* carries parts, this is the service that was provided while loading the part.

SimTalk

[getWorkingWorkersTable \[SimTalk\]](#)

Tools Menu

Tools Menu

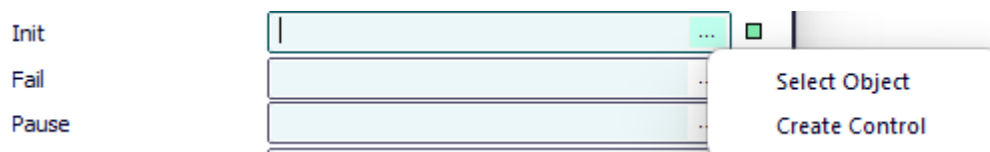
The **Tools Menu** provides these menu commands:

[Edit Controls](#) > [Init control](#)

[Edit Observers](#)

Init Control [WorkerPool]

Modifies the built-in behavior of the object. The object calls the **Init Control** at the start of the simulation run during the *init* phase.



Remarks

The **Init Controls** are called before the events are computed. The normal *init methods* on the other hand are executed after the initial events have been computed.

Within the **Init Control** you can, for example, change the **Workers to Create** table of the *WorkerPool* for the next simulation run. This cannot be accomplished in an *init method* as this method would be called too late in time.

You can also initialize attributes which affect event generation, for example the availability. Here the **Init Controls** are executed before events are computed. The normal *init methods* are executed after the initial events have been computed.

Select the Path to an Existing Method

Click the ellipsis button. Navigate to the location of the *Method* in the dialog **Select Object [for controls]** and click **OK**. This inserts the name of the *Method* into the text box of the **Control**.



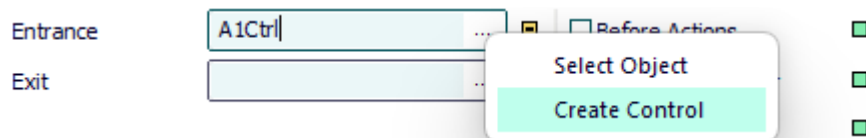
Press **F2** in the text box to open the *Method*. Then type in the source code of the **Control**.

Instead of choosing **Select Object**, you can also select the *Method* in a *Frame*, drag it to the text box and drop it there.

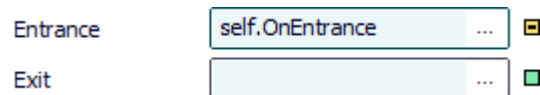
Create a Control That is a Method of the Object

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- Type a meaningful name into the text box and select **Create Control [context menu]**. *Plant Simulation* then inserts `self.Name_you_typed_in_for_the_control`, such as `self.A1Ctrl`.



- Select **Create Control** on the empty text box. *Plant Simulation* then inserts `self.OnBuilt_in_name_of_the_control`, such as `self.OnEntrance`.



Type the source code of this control into the *Method* that opens.

To edit the source code later on:

- Press **F2**.
- Or hold down **Shift** and double-click into the text box.
- Or select **Open Object** on the context menu.
- Or click the tab **User-defined** and double-click the name of the *Method* in the list.
- To delete this control, delete the *user-defined attribute*. If you only delete the name from the text box, the *user-defined attribute* is retained.

SimTalk

InitCtrl [SimTalk] - WorkerPool

See also

Select Object [for controls]

Help Menu

The commands are described under the [Help Menu](#).

Methods of the WorkerPool

_Methods of the WorkerPool

The *WorkerPool* provides:

- The *methods* listed in the table of contents to the left.
- The [Methods of All Objects](#).

To view all of the *methods*, *read-only attributes*, and *attributes* of the object, open the window [Show Attributes and Methods](#). The figure below illustrates the information using the example of the object *Station*.

Name	Value	Inherited	Watchable	Signature
contentsList	[]			([Contents:table]) -> any
createAttr				(NameOfUserDefinedAttribute:string, DataType:string) -> boolean
createIcon				(IconName:string, Width:integer, Height:integer) -> integer
deleteAttr				(NameOfUserDefinedAttribute:string) -> boolean

Name of the method Identifier and data type of the parameter you enter Data type of the return value

Name	Value	Inherited	Watchable	Signature
moveToFolder				(Folder:object) -> object
MU	VOID			([Index:integer:=1]) -> object
MUPart	VOID			([Index:integer:=1]) -> object

Name of the method Identifier and data type of the parameter you enter Default value Data type of the return value

- Select **Show Attributes and Methods** on the context menu of the **Class Library** to show the *methods*, *read-only attributes*, and *attributes* of the selected [Class \[general description\]](#).
- Press the **F8** key or click **Show Attributes and Methods** on the **Home** ribbon tab of the *Frame* into which you inserted an instance to show the *methods*, *read-only attributes*, and *attributes* of the selected [Instance \[general description\]](#).